



Master in Computer Science
Software Engineering

Cloud Computing

Report 6

Identity and Access Management (IAM)

Team Members

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Part 1. IAM Policies

Task 1. Create Policy Number 1

We create Policy Number 1 which allows:

- Access S3 completely
- Only create EC2 instances
- Full access to RDS

The policy is created in JSON format:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeInstances",
        "ec2:StartInstances",
        "rds:*",
        "s3:*",
        "ec2:CreateTags",
        "ec2:DescribeInstanceAttribute",
        "ec2:DescribeRegions",
        "ec2:DescribeInstanceTypes",
        "ec2:RunInstances",
        "ec2:AssociateAddress",
        "ec2:DescribeInstanceStatus"
      ],
      "Resource": "*"
    }
  ]
}
```

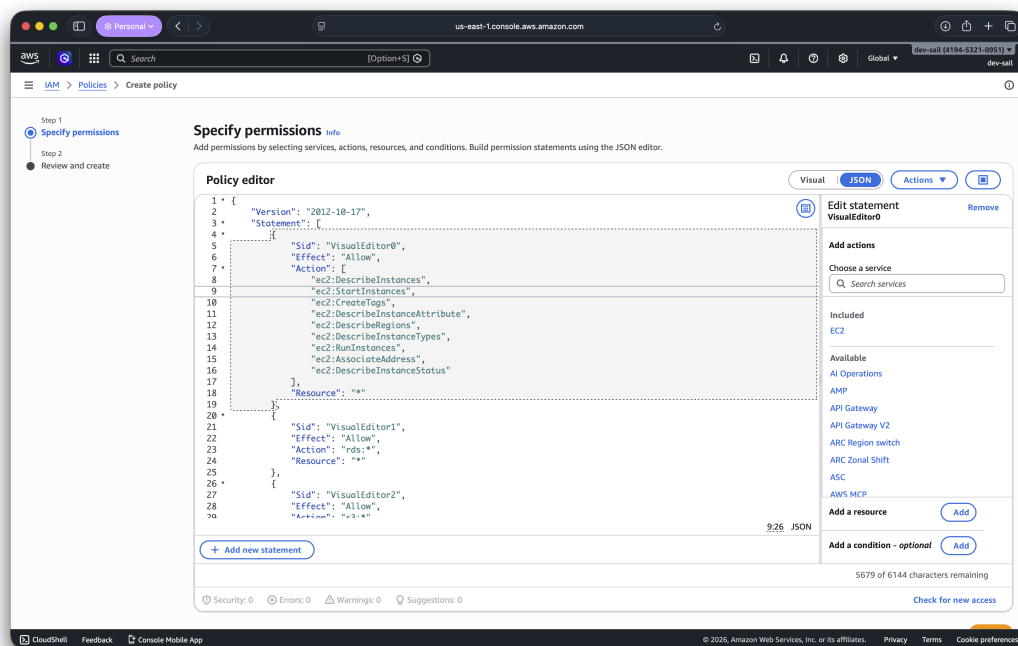


Figure 1: Policy 1 JSON configuration in AWS console

Task 2. Create Policy Number 2

We create Policy Number 2 which allows:

- Access CloudWatch and billing completely
- Can only list EC2 and S3 resources

The policy is created in JSON format:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeInstances",
        "s3:ListAllMyBuckets",
        "cloudwatch:*",
        "s3:ListBucket"
      ],
      "Resource": "*"
    }
  ]
}
```

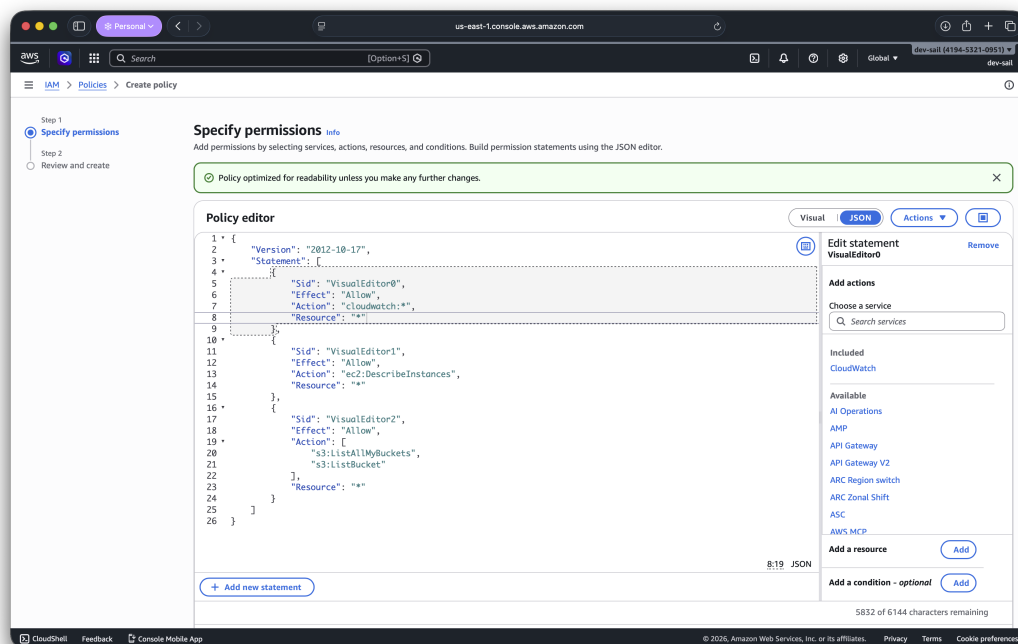


Figure 2: Policy 2 JSON configuration in AWS console

Task 3. Attach Policy Number 1 to Dev Team

We attach Policy Number 1 to the Dev Team group.

Steps performed:

1. Navigate to IAM → User groups
2. Select the DevTeam group
3. Click on the Permissions tab

4. Click Add permissions → Attach policies
5. Search for and select Policy1
6. Click Attach policies

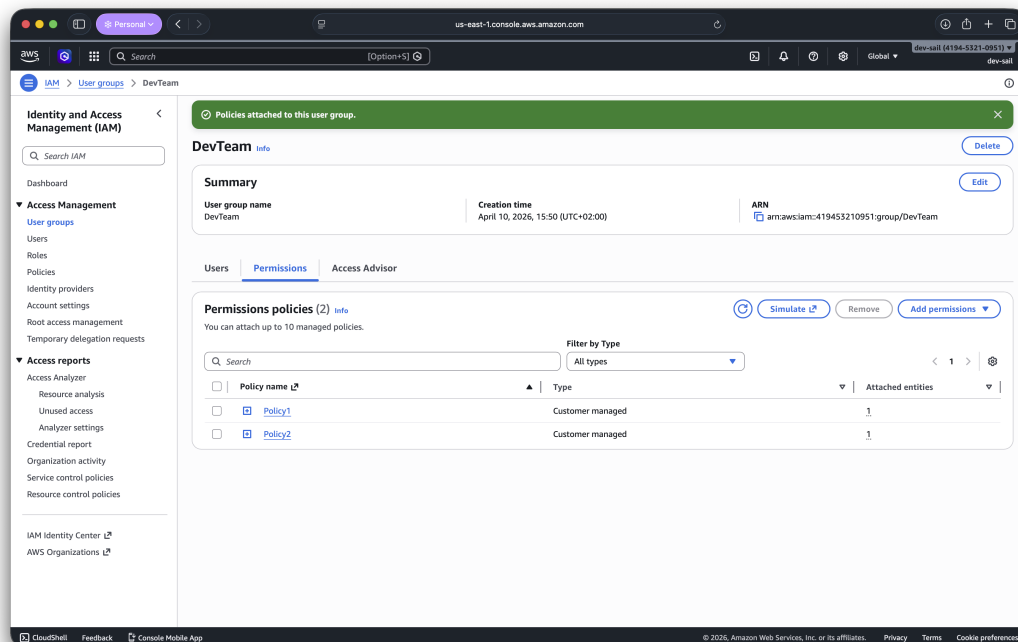


Figure 3: Attaching Policy 1 to DevTeam group

Task 4. Attach Policy Number 2 to Ops Team

We attach Policy Number 2 to the Ops Team group.

Steps performed:

1. Navigate to IAM → User groups
2. Select the OpsTeam group
3. Click on the Permissions tab
4. Click Add permissions → Attach policies
5. Search for and select Policy2
6. Click Attach policies

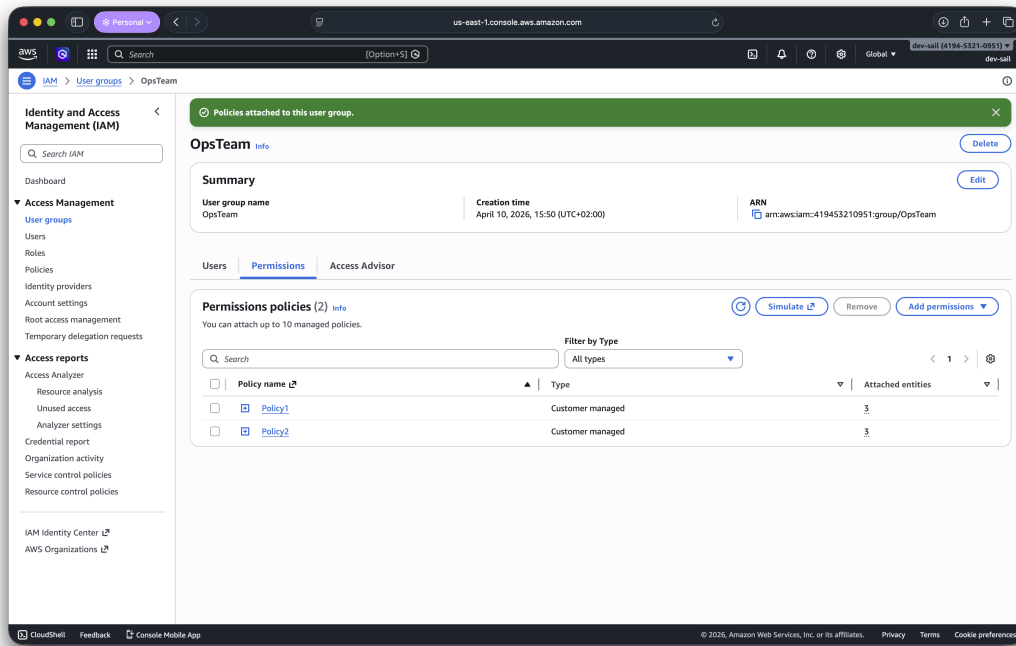


Figure 4: Attaching Policy 2 to OpsTeam group

Part 2. IAM Assignment

Task 1. Create IAM Users

We create 4 IAM users: Dev1, Dev2, Test1, and Test2.

Steps performed:

1. Navigate to IAM → Users
2. Click **Create user**
3. Enter the username (Dev1, Dev2, Test1, or Test2)
4. Proceed through the user creation wizard
5. Complete the user creation process

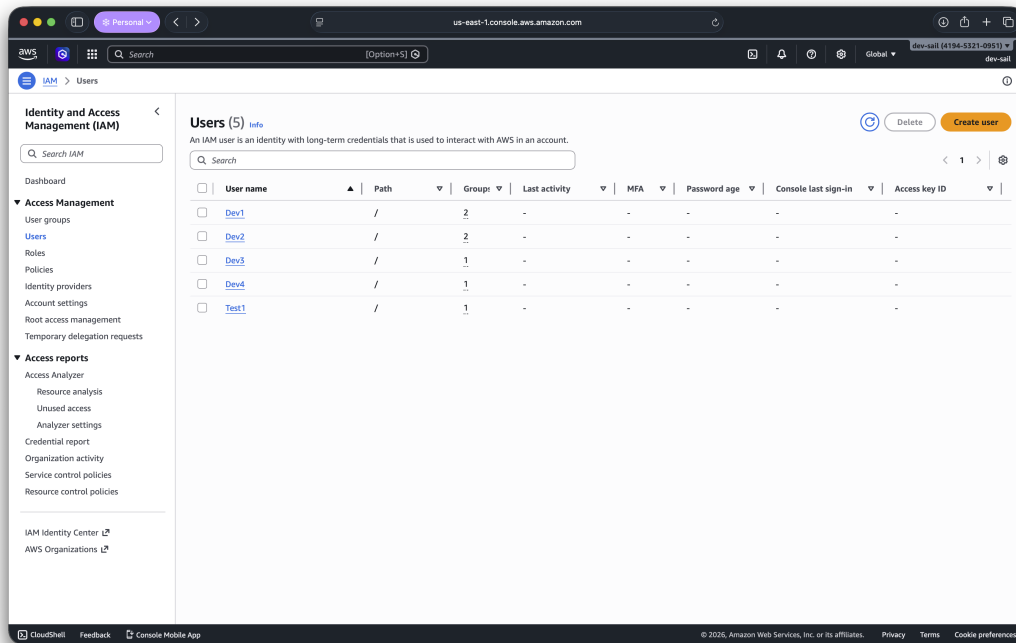


Figure 5: IAM Users dashboard showing created development and testing users

Task 2. Create User Groups

We create 2 groups: Dev Team and Ops Team.

Steps performed:

1. Navigate to IAM → User groups
2. Click **Create group**
3. Enter the group name
4. (Optional) Attach policies during group creation
5. Complete the group creation

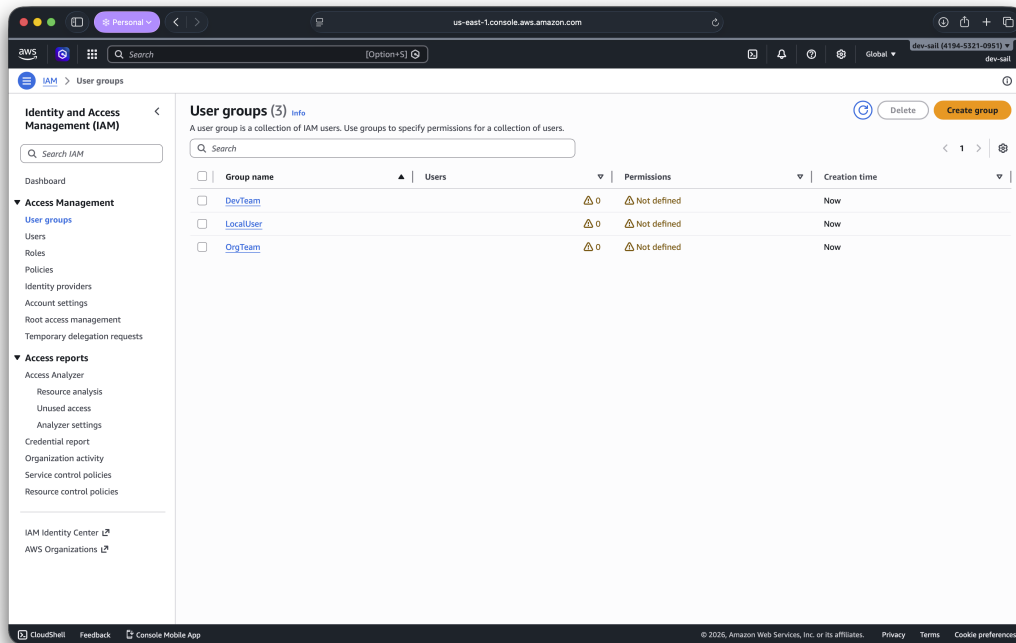


Figure 6: User groups dashboard showing DevTeam, LocalUser, and OpsTeam groups

Task 3. Add Dev1 and Dev2 to Dev Team

We add Dev1 and Dev2 to the Dev Team group.

Steps performed:

1. Navigate to IAM → User groups
2. Select the DevTeam group
3. Click Add user to group
4. Select Dev1 and Dev2
5. Click Add users

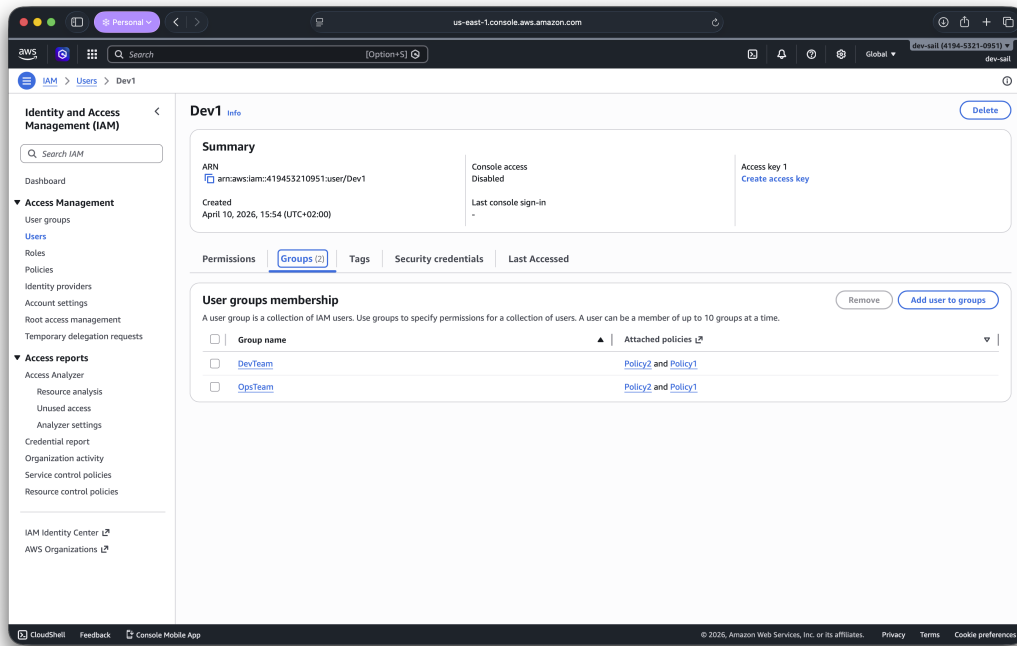


Figure 7: Dev1 user group membership showing association with DevTeam and OpsTeam

Task 4. Add Dev1, Test1, and Test2 to Ops Team

We add Dev1, Test1, and Test2 to the Ops Team group.

Steps performed:

1. Navigate to IAM → User groups
2. Select the OpsTeam group
3. Click Add user to group
4. Select Dev1, Test1, and Test2
5. Click Add users

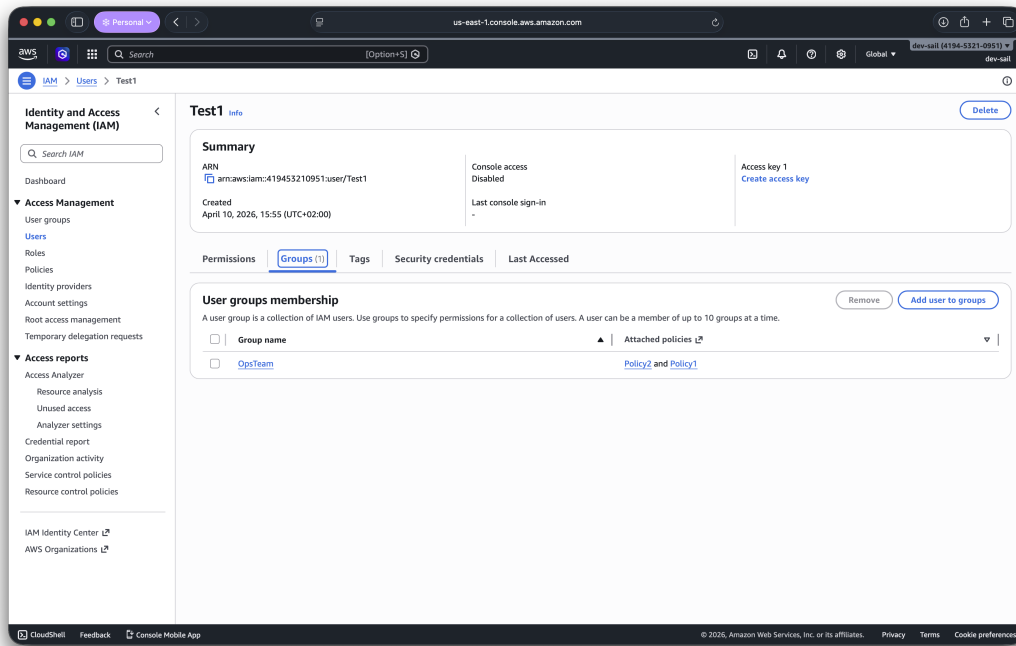


Figure 8: Test1 user group membership showing association with OpsTeam group

Part 3. Conclusion

In this lab, we accomplished the following tasks:

Part 1 - IAM Policies:

- Created Policy1 with S3 full access, EC2 instance creation, and RDS full access
- Created Policy2 with CloudWatch and billing full access, EC2 and S3 list-only access
- Attached Policy1 to the Dev Team group
- Attached Policy2 to the Ops Team group

Part 2 - IAM Assignment:

- Created 4 IAM users (Dev1, Dev2, Test1, Test2)
- Created 2 groups (Dev Team and Ops Team)
- Added Dev1 and Dev2 to the Dev Team group
- Added Dev1, Test1, and Test2 to the Ops Team group